

The Greedy Algorithm

Problem: Given $M = (E, \mathcal{I})$, and given
a weight function $w: E \rightarrow \mathbb{R}$, find
 $I \in \mathcal{I}$ maximizing $W(I) = \sum_{e \in I} w(e)$

Greedy Alg: Order the elements so $w_1 \geq w_2 \geq \dots \geq w_{|E|}$

$I \leftarrow \emptyset$

At i 's iteration, if $I + e \in \mathcal{I}$ then $I \leftarrow I + e$

return I .

Thm: The Greedy alg. returns a max weight base $\forall w \geq 0$
iff M is a matroid.

Proof — 1/4

Assume M is a matroid. Let trajectory of alg. be: $\phi = I_0 \subseteq I_1 \subseteq I_2 \subseteq \dots \subseteq I_{|E|}$

Note: $|I_k| \leq k$.

We show: at any iteration k , $\exists$ opt solution - i.e., $(*)$

a max weight base, B_k st: $I_k \subseteq B_k$
 $B_k \setminus I_k \subseteq \{k+1, \dots, n\}$ ($n = |E|$).

We prove by induction on k . $k=0$, $I_0 = \phi$ so prop. holds.

Assume this is true up to iteration $k-1$.

I_{k-1} , $\exists B_{k-1}$ with $I_{k-1} \subseteq B_{k-1}$ and $B_{k-1} \setminus I_{k-1} \subseteq \{k, \dots, n\}$.

What does Greedy do at iteration k ?

- If Alg. does not take elt k it must be that $I_{k-1} + k \notin \mathcal{I}$

$\Rightarrow k \notin B_{k-1} \Rightarrow \underline{B_k} \triangleq B_{k-1}$, B_k satisfies induction.

Proof — 2/4

• If Alg does take k : $I_k = I_{k-1} + k$

— If $k \in B_{k-1} \Rightarrow I_k \subseteq B_{k-1}$, so again : $B_k \triangleq B_{k-1}$.

— If $k \notin B_{k-1}$: $I_k \in \mathcal{I}$, $B_{k-1} \in \mathcal{I}$

we can extend I_k to a base by taking elts

of B_{k-1} . Call this B' . $k \in B'$, $k \notin B_{k-1}$

We know: $B' \setminus B_{k-1} = k$, $j \in B_{k-1} \setminus B'$

$$B_{k-1} \setminus I_{k-1} \subseteq \{k+1, \dots, n\} \Rightarrow j \in \{k+1, \dots, n\}$$

$$\Rightarrow w_j \leq w_k$$

$$\Rightarrow \underline{w(B')} = w(B_{k-1}) + w_k - w_j \geq \underline{w(B_{k-1})}$$

$\Rightarrow B'$ is an optimal base that satisfies

required properties: $I_k \subseteq B'$, $B' \setminus I_k \subseteq \{k+1, \dots, n\}$

Take: $B_k = B'$.

Proof — 3/4

Suppose that $(E, \mathcal{I})$ that is not a matroid,
i.e., $\mathcal{I}$ violates one of two matroid properties,

① Suppose $\mathcal{I}$ not downward closed:

$\exists S \subseteq T, T \in \mathcal{I}$ but $S \notin \mathcal{I}$.

Choose: $w_i = \begin{cases} 2 & i \in S \\ 1 & i \in T \setminus S \\ 0 & \text{o.w.} \end{cases}$

Greedy alg will skip at least one elt of
 S , while the opt solution
is all of T .

Proof — 4/4

Suppose $\mathcal{L}$ is downward closed but fails

the extension property: $\exists S, T \in \mathcal{L}$,

with $|S| < |T|$, but $S \cup \{i\} \notin \mathcal{L} \quad \forall i \in T \setminus S$.

Choose:

$$w_i = \begin{cases} 1 + \frac{1}{2|S|} & i \in S \\ 1 & i \in T \setminus S \\ 0 & \text{o.w.} \end{cases}$$

Greedy alg returns S :

$$w(S) = |S| \left(1 + \frac{1}{2|S|} \right) = |S| + \frac{1}{2}$$

$$w(T) \geq |T| \geq |S| + 1.$$

